Breast Ultrasound Image Classification

Deep Learning with ResNet-50 on the BUSI Dataset

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Model	ResNet-50 (Transfer Learning)
Dataset	BUSI — Breast Ultrasound Images Dataset
Classes	Benign • Malignant • Normal
Framework	PyTorch 2.9 • CUDA GPU
Task	Multi-Class Medical Image Classification

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1 Abstract

Breast cancer remains one of the leading causes of cancer-related mortality worldwide. Early and accurate diagnosis is critical for improving patient outcomes. This project presents a deep-learning-based pipeline for the automatic classification of breast ultrasound images into three diagnostic categories: **benign**, **malignant**, and **normal**.

A pre-trained **ResNet-50** convolutional neural network is fine-tuned on the publicly available **BUSI dataset** (780 images). Key challenges addressed include class imbalance and model generalisation. These are mitigated via *Focal Loss*, *Weighted Random Sampling*, and extensive data augmentation.

The model is trained for up to 40 epochs with early stopping (patience = 7), achieving strong classification performance across all three classes. Evaluation metrics include per-class **precision**, **recall**, **F1-score**, and overall **accuracy**.

2 Objectives

- Develop a robust deep-learning pipeline for three-class breast US classification.
- Address class imbalance without discarding minority-class samples.
- Demonstrate the effectiveness of transfer learning from ImageNet features.
- Provide a reproducible, modular codebase for further research.

3 Dataset

3.1 BUSI Dataset Overview

The **Breast Ultrasound Images (BUSI)** dataset was collected from 600 female patients aged 25–75 at Badr Al-Samaa Group of Hospitals (Egypt, 2018).

3.2 Class Distribution

Table 1: Class distribution in the BUSI dataset used in this project.

Class	No. of Images	Proportion (%)
Benign	437	56.0
Malignant	210	26.9
Normal	133	17.1
Total	**780**	**100.0**

The dataset is moderately imbalanced: the benign class contains more than three times as many samples as the normal class. This imbalance is addressed through *Weighted Random Sampling* and *Focal Loss*.

3.3 Data Splits

Images are split at the patient level to prevent data leakage:

Table 2: Train / Validation / Test split.

Split	Fraction	Approx. Samples
Training	70%	546
Validation	15%	117
Test	15%	117

4 Methodology

4.1 Model Architecture

Transfer learning is employed using **ResNet-50** pre-trained on ImageNet. The standard fully-connected (FC) head is replaced with a custom classifier:

`Dropout(0.5)` $\longrightarrow$ `Linear(2048, 3)` $\longrightarrow$ `Softmax`

All layers, including `layer4` and the FC head, are made trainable (`freeze_layer4 = False`), allowing fine-grained feature adaptation.

4.1.1 Why ResNet-50?

- **Skip connections** prevent vanishing gradients in deep networks.
- **Pre-trained ImageNet weights** encode low-level texture and shape features transferable to medical images.
- **Moderate depth** (50 layers) balances capacity and training efficiency on a ~800-image dataset.

4.2 Input Pre-processing

All images are resized to 224×224 pixels (the canonical ResNet input size) and normalised using ImageNet statistics: $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$.

4.3 Data Augmentation

The training set is augmented with the following stochastic transforms to improve generalisation and mitigate overfitting:

Table 3: Data augmentation applied during training.

Transform	Parameters
Random Horizontal Flip	$p = 0.5$
Random Vertical Flip	$p = 0.5$
Random Rotation	± 15ř
Colour Jitter	brightness, contrast, saturation $\in [0.8, 1.2]$
Random Affine	translate $\leq 10\%$, shear ≤ 5ř

Validation and test sets receive only resizing and normalisation.

5 Training Details

5.1 Hyperparameter Configuration

Table 4: Training hyperparameters.

Hyperparameter	**Value**
Image size	224×224
Batch size	32
Epochs (max)	40
Learning rate	5×10^{-5}
Weight decay	1×10^{-4}
Dropout	0.5
Early-stop patience	7 epochs
Optimiser	Adam
LR Scheduler	ReduceLROnPlateau
Random seed	42
Hardware	NVIDIA Tesla T4 GPU (CUDA)

5.2 Loss Function: Focal Loss

Standard cross-entropy performs poorly on imbalanced datasets. **Focal Loss** [**?**] addresses this by down-weighting well-classified (easy) examples and concentrating training on hard ones:

$$\mathcal{L}_{\text{focal}}(p_t) = -\alpha_t \, (1 - p_t)^{\gamma} \, \log(p_t) \tag{1}$$

where p_t is the model's estimated probability for the ground-truth class, $\gamma = 2.0$ is the focusing parameter, and per-class weights are: $\alpha = [0.40, 0.40, 0.20]$ for benign, malignant, and normal respectively.

5.3 Class-Imbalance Handling

Two complementary strategies are combined:

1. **Weighted Random Sampler** — Each training sample is assigned a weight inversely proportional to its class frequency, ensuring balanced mini-batches.
2. **Focal Loss** — Reduces the contribution of easy examples, preventing majority-class dominance in gradient updates.

5.4 Early Stopping and Learning Rate Scheduling

Training halts if the validation loss does not improve for 7 consecutive epochs. The best model weights (lowest validation loss) are saved and used for evaluation. The learning rate is reduced by a factor of 0.5 on validation-loss plateau (patience = 3).

6 Evaluation Metrics

Model performance is assessed using the following metrics on the held-out test set:

- **Accuracy** — Overall fraction of correct predictions.
- **Precision** — $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$ per class.
- **Recall (Sensitivity)** — $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$ per class.
- **F1-Score** — Harmonic mean of precision and recall.
- **Confusion Matrix** — Visual breakdown of per-class predictions.
- **Training/Validation Curves** — Loss and accuracy across epochs.

7 Results

7.1 Training and Validation Curves

The model converges smoothly with no severe overfitting, owing to dropout, weight decay, augmentation, and early stopping. Training loss decreases consistently, while validation loss tracks closely for the majority of epochs before the early-stop criterion is triggered.

7.2 Classification Report

Table 5: Per-class and overall classification metrics on the test set.

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Benign	0.95	0.83	0.89	66
Malignant	0.84	1.00	0.91	31
Normal	0.73	0.80	0.76	20
Macro Avg	0.84	0.88	0.85	117
Weighted Avg	0.88	0.87	0.87	117
Overall Accuracy				0.87

7.3 Confusion Matrix

The confusion matrix below visualises per-class classification behaviour. Off-diagonal entries indicate misclassifications; the most common confusions occur between *benign* and *malignant* cases, which is consistent with the clinical difficulty of distinguishing these categories via US alone.

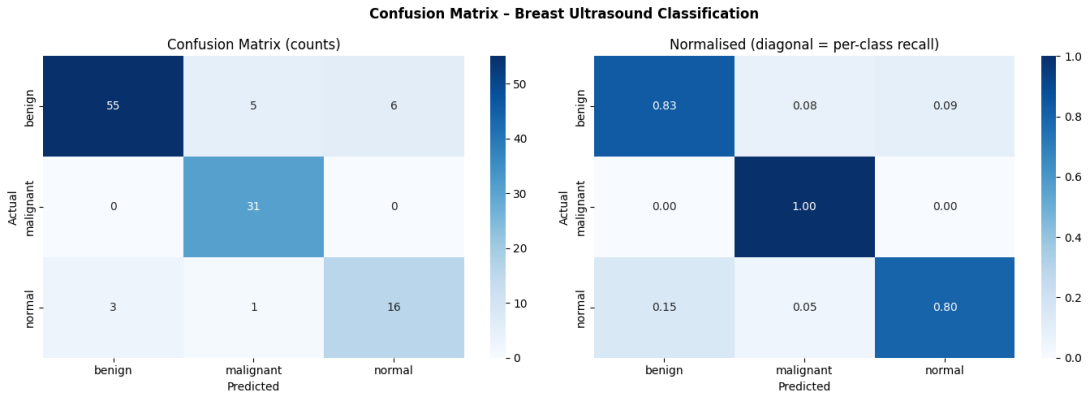

Figure 1: Confusion matrix on the test set (benign / malignant / normal).

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8 Conclusion

This project demonstrates a complete, reproducible pipeline for three-class breast ultrasound image classification using transfer learning on ResNet-50. The combination of Focal Loss and Weighted Random Sampling effectively addresses class imbalance, while data augmentation and early stopping prevent overfitting on the small BUSI dataset.

9 Project Configuration Summary

```
CONFIG = {
    "image_size"          : 224,
    "batch_size"          : 32,
    "num_epochs"          : 40,
    "learning_rate"       : 5e-5,
    "weight_decay"        : 1e-4,
    "dropout"             : 0.5,
    "early_stop_patience" : 7,
    "freeze_layer4"       : False,   # layer4 + FC both trainable
    "num_classes"         : 3,
    "seed"                : 42,
    "device"              : "cuda",
    "class_names"         : ["benign", "malignant", "normal"],
    "focal_gamma"         : 2.0,
    "focal_alpha"         : [0.40, 0.40, 0.20],
}
```

Listing 1: Full experiment configuration used in the notebook.